

SQL fundamentals

SELECT & Filtering

Selecting & Sorting

1. **SELECT ***

FROM employees;

employees

Id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-01	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. **SELECT DISTINCT** department

FROM employees;

employees

department
IT
HR
Finance
Marketing

3. **SELECT** first-name,

last-name

FROM employees

ORDER BY Salary **DESC**

employees

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. **SELECT** id,
first_name,
last_name,
Salary
FROM employees
ORDER BY Salary **DESC**
LIMIT 3;

employees

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. **SELECT** id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

employees

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. **SELECT** id, first_name, last_name, department, Salary
FROM employees
WHERE department = 'finance' **AND** Salary > 60000;

employees

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

7. **SELECT** id, first_name, last_name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';

employees

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. **SELECT** id, first_name, last_name, department
FROM employees
WHERE **NOT** department = 'IT';

employees

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. **SELECT**
FROM
WHERE

9. **SELECT** id,
first-name,
last-name,
department

WHERE

FROM employees

WHERE department **IN** ('IT', 'HR', 'Finance');

employees

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. **SELECT** id,
first-name,
last-name,
department,
City

FROM
WHERE

employees

WHERE department = 'IT'

AND Salary > 65000

AND city = 'Johannesburg';

employees

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg